LEONARDO SANTOS

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Homepage ◇ Google Scholar ◇ Github ◇ LinkedIn

EDUCATION

KTH Royal Institute of Technology *August 2025-Present, Sweden*
Ph.D Student in the Division of Robotics Perception and Learning (RPL)
Supervisor: Prof. Iolanda Leite, Co-supervisor: Prof. Jana Tumova.

UFMG Federal University of Minas Gerais *July 2025, Brazil*
B.Sc. in Electrical Engineering, Emphasis in Robotics
GPA: 86/100

SKILLS

Programming	Python, C++, MATLAB
Technologies	ROS, RViz, Git, LaTeX
Languages	Portuguese (native), English (fluent)

RESEARCH EXPERIENCE

Graduate Research Assistant August 2025 – Present
KTH | *Prof. Iolanda Leite* | *RPL* Stockholm, Sweden

· Doing research on Interpretable Reinforcement Learning for applications in Human-Robot Interaction.

Robotics Institute Summer Scholar (RISS) June 2024 – August 2024
Carnegie Mellon University | *Prof. Andrea Bajcsy* | *Intent Lab* Pittsburgh, USA

· At Intent Lab, I worked on a method to update safety specifications online from human language feedback using a Vision Language Model, and apply these safety specifications in a downstream reachability controller for safe navigation. We demonstrated the system in both hardware and simulation experiments [1].

· More information and demos on the project website.

Robotics Institute Summer Scholar (RISS) June 2023 – August 2023
Carnegie Mellon University | *Prof. Sebastian Scherer* | *AirLab* Pittsburgh, USA

· At AirLab, I worked on a framework based on Random Finite Sets that unifies search and track optmization objectives in informative path planning. I contributed to both theoretical and implementation developments [2].

Research Assistant February 2021 – May 2023
UFMG | *Prof. Douglas Macharet* | *VeRLab* Belo Horizonte, Brazil

· At VeRLab, I worked on multi-robot system for warehouse management [3], and on communication-aware path planning for multi-robot systems [4].

PUBLICATIONS

[1] L. Santos*, Z. Li*, L. Peters, S. *Bansal*†, and A. *Bajcsy*†, "Updating robot safety representations online from natural language feedback," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[2] L. Santos*, B. Moon*, S. Scherer, and H. Van Nguyen, "Unisat: Unified-objective belief model and planner to search for and track multiple objects," in *AIAA SCITECH 2025 Forum*, 2025, p. 2114.

[3] C. C. G. Ribeiro, L. H. M. C. dos Santos, and D. G. Macharet, "Collaborative ugv/uav path planning for inventory management in warehouses," in *2022 Latin American Robotics Symposium (LARS)*, 2022, pp. 121–126.

[4] L. H. Dos Santos, C. C. Ribeiro, and D. G. Macharet, "Communication backbone deployment with connectivity maintenance," in *2026 Brazilian Conference on Robotics (CROS)*, IEEE, 2026, pp. 1–6.

[5] H. Kuehn, L. Santos, and I. Leite, "Clarifying constraints in interactive robot learning with language feedback," in *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*, 2026, pp. 816–824.

[6] A. A. Neto, V. C. d. S. Campos, L. Santos, and D. G. Macharet, "Minimal exposure path problem for multi-agent cyber-physical systems," *Robotics and Autonomous Systems*, p. 105 638, 2026.

* Indicates equal contribution, † indicates equal advising.

AWARDS, GRANTS AND SCHOLARSHIPS

KTH International Collaboration Grant to carry out a research project in collaboration with researchers from IP Paris, France (100,000 SEK, 10,000 EUR) *Spring 2026*
KTH General Travel Foundations Grant to attend RLSS 2026 (18,000 SEK, 1,800 EUR) *Summer 2026*
IEEE RAS Travel Grant to attend ICRA 2025 *Spring 2025*
Google Undergraduate Research Scholarship, 1 of 3 scholars at CMU *Summer 2024*
Minas Mundi Study Abroad, selected UFMG student *Fall 2023*
Mobility21 Undergraduate Research Scholarship, 1 of 3 scholars at CMU *Summer 2023*
National Finalist, top 3 Overall, 2nd SARC-BARINet Aerospace Competition *Fall 2022*
Top 15 Papers Award out of 80 submissions, LARS 2022 *Fall 2022*
Minas Gerais Research Support Foundation Scholarship Fall 2022, Spring 2023
Brazilian National Council for R&D Scholarship Fall 2021, Spring 2022

SERVICE AND OUTREACH

Reviewer July 2025 – Present

· NeurIPS 2026; IROS 2026; ACM/IEEE HRI 2026; NeurIPS 2025 (Position paper track).

Member of Admissions Committee January 2025
Carnegie Mellon University's Robotics Institute Summer Scholars

· Reviewed applications for the Robotics Institute Summer Scholars (RISS) program at Carnegie Mellon University (CMU).

· RISS is a funded summer research program that brings national and international students to do research at the CMU Robotics Institute.

Co-Lead Organizer June 2023 – August 2024
RoboLaunch Pittsburgh, USA

- Helped organize a robotics seminar series aimed at introducing undergraduate students to robotics through a series of accessible talks from speakers from academia and industry.
- Led logistics involving speaker invitations, advertisement, and general team coordination.
- During the time I was involved, we hosted speakers from CMU, MIT, the University of Washington, the University of Adelaide, ETH Zurich, among others.

Co-Founder and Webmaster September 2024 – July 2025
UFMG IEEE Student Branch Belo Horizonte, Brazil

- Organized events to promote and advertise research at UFMG, including seminars on microelectronics, AI, robotics, and biomedical engineering; research partnerships between Brazilian and French universities.
- Was responsible for everything related to branding, external communications, and social media presence.

SOCIETY MEMBERSHIPS

IEEE RAS Student Member *2025 – Present*
IEEE Student Member *2024 – Present*